

Vote, or Search a Tree

DATA 443/643 · Week 13, note 13.2

Week 13

American University

Ahmad Mousavi

Ideas from Stanford CS224N 2026 L12 (SC +17.9 pp GSM8K). Original slides; that course is not copied.

Lecture plan

- 1 k traces, majority on the parsed answer
- 2 Wang et al.: +17.9 pp on GSM8K
- 3 ToT scores partial nodes
- 4 A shared bug still wins the vote

Key goal: vote on answers. $k=1$ is greedy CoT.

First time: what / why / formula

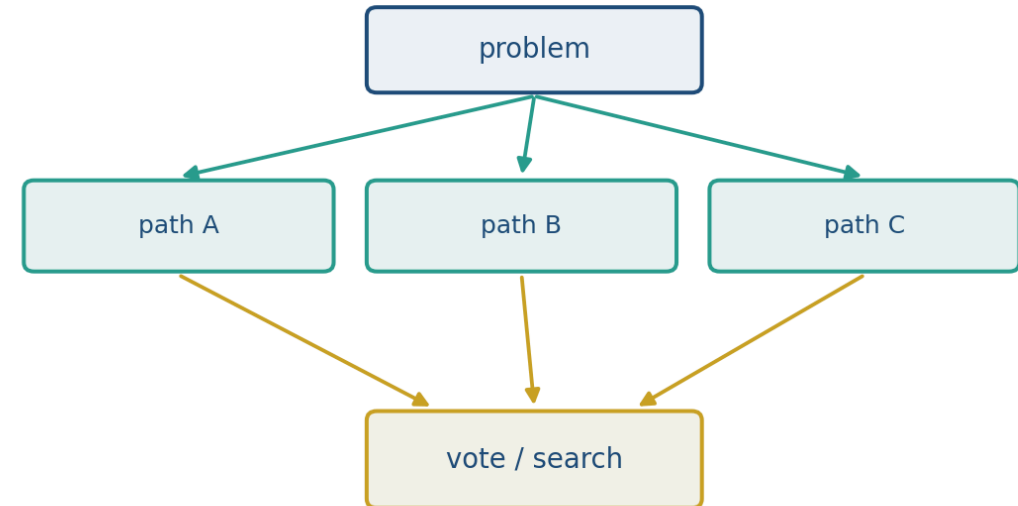
- What / why. SC: sample k traces (temperature > 0), parse answers, vote. ToT: expand nodes, score, prune, expand again. One CoT is noisy. Voting on the answer (not the wording) lifts GSM8K-style math in the CS224N L12 figure ($\sim +17.9$ pp in that citation). ToT helps when you need a tree, not k independent tapes.
- Architecture. SC: decoder + parser + vote. ToT: a frontier of partial strings + a scorer (LM or checker) + beam/DFS.
- Formula. SC: $\hat{y} = \text{mode}\{\text{parse}(\tau_1), \dots, \text{parse}(\tau_k)\}$. ToT has no single loss; it is search.
- Tradeoffs. + SC is simple. – $k \times$ cost; shared bugs survive the vote; ToT needs a scorer (math not in this hour); ReAct (14.2) calls tools instead of searching thoughts.

How. Temperature 0, $k = 3$ is not SC (identical greedy traces). Vote on the parsed number.

Test-time compute as k forwards

- Temperature > 0 . Parse each box. Majority.
- CS224N L12: extra tokens are extra compute. SC spends k of them.
- Do not assign DeepSeek-R1 / GRPO as homework.

Self-consistency votes; Tree-of-Thoughts searches partial steps



Majority on the parsed answer

$$\hat{a} = \operatorname{argmax}_a |\{i: \text{parse}(y_i) = a\}|$$

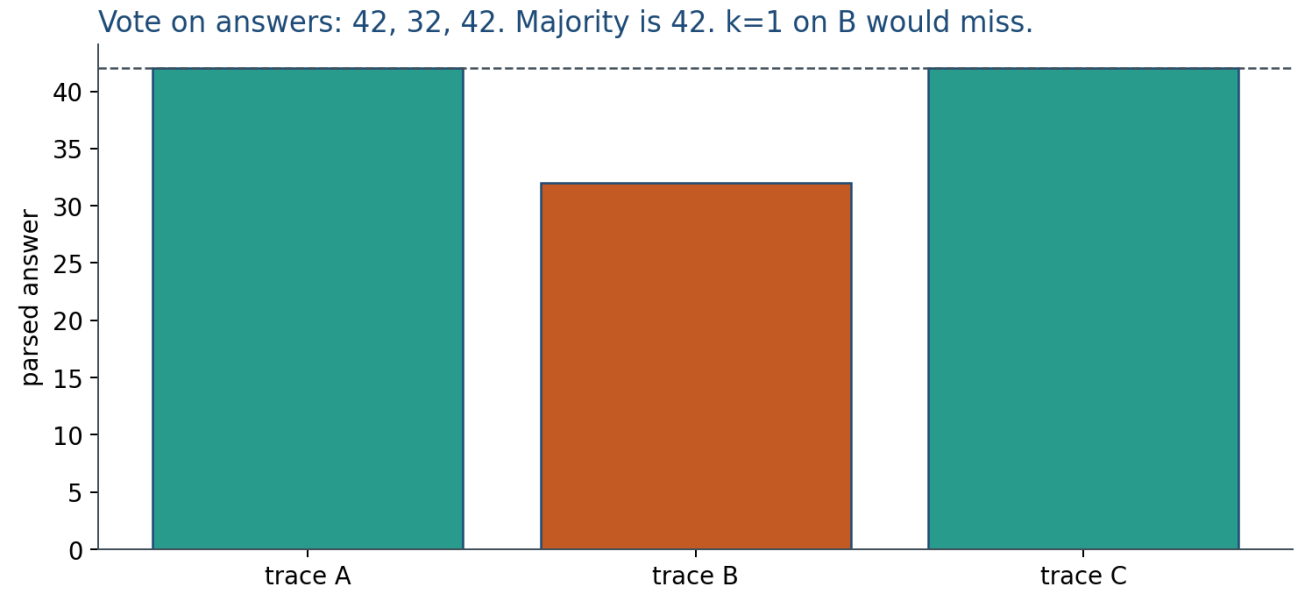
Vote on the number, not on the wording of the steps.

Tie-break is a method choice. Write it down.

Temperature 0 makes all k traces the same.

Lab 13 numbers, then GSM8K

- $23+19$: answers 42, 32, 42. Majority 42.
- $k = 1$ on B scores 32 and misses.
- Wang et al., cited in CS224N L12: +17.9 pp vs greedy CoT.



Two ways to spend the budget

Self-consistency

i.i.d. full traces. Vote. Shared formula bugs survive.

Tree-of-Thoughts

Expand partial thoughts, score, prune. ReAct (Week 14) calls a tool instead.

What to take from this lecture

Plot k

Accuracy vs k, plus greedy CoT at k=1.

GSM8K punch

+17.9 pp is a lift, not a proof of faithful steps.

Lab 13

Vote on constructed answers. Then flag 13.3.